

Motor	xxx
Turns	22
Termination ("D" or "Y")	d
Kv	447
Wire Length	68
Milli ohm/ft	10.126
	.

Load data, only y
Turn calculator.

RPM/V
inch

Block 1

Turns	Delta Kv	Star/Y Kv	Wire Length (in)	resist wire milliohm	Rm resist Delta	Rm resist Star
1	9834	5678	15.1	12.73	8.49	25.47
2	4917	2839	18.2	15.34	10.23	30.68
3	3278	1893	21.3	17.95	11.97	35.90
4	2459	1419	24.4	20.56	13.71	41.12
5	1967	1136	27.5	23.17	15.44	46.33
6	1639	946	30.5	25.78	17.18	51.55
7	1405	811	33.6	28.38	18.92	56.77
8	1229	710	36.7	30.99	20.66	61.98
9	1093	631	39.8	33.60	22.40	67.20
10	983	568	42.9	36.21	24.14	72.42
11	894	516	46.0	38.82	25.88	77.63
12	820	473	49.1	41.42	27.62	82.85
13	756	437	52.2	44.03	29.36	88.07
14	702	406	55.3	46.64	31.09	93.28
15	656	379	58.4	49.25	32.83	98.50
16	615	355	61.5	51.86	34.57	103.71
17	578	334	64.5	54.47	36.31	108.93
18	546	315	67.6	57.07	38.05	114.15
19	518	299	70.7	59.68	39.79	119.36
20	492	284	73.8	62.29	41.53	124.58
21	468	270	76.9	64.90	43.27	129.80
22	447	258	80.0	67.51	45.00	135.01
23	428	247	83.1	70.11	46.74	140.23
24	410	237	86.2	72.72	48.48	145.45
25	393	227	89.3	75.33	50.22	150.66
26	378	218	92.4	77.94	51.96	155.88
27	364	210	95.5	80.55	53.70	161.10
28	351	203	98.5	83.16	55.44	166.31
29	339	196	101.6	85.76	57.18	171.53
30	328	189	104.7	88.37	58.91	176.74
31	317	183	107.8	90.98	60.65	181.96
32	307	177	110.9	93.59	62.39	187.18
33	298	172	114.0	96.20	64.13	192.39
34	289	167	117.1	98.81	65.87	197.61
35	281	162	120.2	101.41	67.61	202.83
36	273	158	123.3	104.02	69.35	208.04
37	266	153	126.4	106.63	71.09	213.26
38	259	149	129.5	109.24	72.83	218.48
39	252	146	132.5	111.85	74.56	223.69
40	246	142	135.6	114.45	76.30	228.91
41	240	138	138.7	117.06	78.04	234.13

Block 2

2Group parallel Delta	2Group parallel Star	Wire Length (in)
19668	11355	7.5
9834	5678	9.1
6556	3785	10.6
4917	2839	12.2
3934	2271	13.7
3278	1893	15.3
2810	1622	16.8
2459	1419	18.4
2185	1262	19.9
1967	1136	21.5
1788	1032	23.0
1639	946	24.5
1513	873	26.1
1405	811	27.6
1311	757	29.2
1229	710	30.7
1157	668	32.3
1093	631	33.8
1035	598	35.4
983	568	36.9
937	541	38.5
894	516	40.0
855	494	41.5
820	473	43.1
787	454	44.6
756	437	46.2
728	421	47.7
702	406	49.3
678	392	50.8
656	379	52.4
634	366	53.9
615	355	55.5
596	344	57.0
578	334	58.5
562	324	60.1
546	315	61.6
532	307	63.2
518	299	64.7
504	291	66.3
492	284	67.8
480	277	69.4

Slat Area
Block 4

Side1	6.9	mm
Side2	6.9	mm
Height1	3	mm
Height2	1	mm
Surface formula		
2	6.9	13.8 mm2

Max turns
Block 5

AWG	Ft Milliohm	D mm	Surface MM2	Turns max	mm2	Real Turns	Turns mm2	% Cooper Paqui.
17	5.054	1.2065	1.1433	4.22	4.8252	3.8	4.34439	62.9622
18	6.386	1.077	0.911	5.30	4.8252	10	9.11008	132.03
19	8.046	0.9703	0.7394	6.53	4.8252	5.25	3.88206	56.2617
20	10.126	0.8661	0.5892	8.19	4.8252	5.25	3.09305	44.8267
21	12.766	0.7772	0.4744	10.17	4.8252	7	3.32089	48.1288
22	16.2	0.6934	0.3776	12.78	4.8252	5.25	1.98252	28.7322
23	20.302	0.6198	0.3017	15.99	4.8252	5.25	1.58399	22.9564
24	25.666	0.5537	0.2408	20.04	4.8252	12	2.88949	41.8767
25	32.363	0.4851	0.1848	26.11	4.8252	17	3.14197	45.5358
26	41.016	0.4318	0.1464	32.95	4.8252	20	2.92878	42.446
30	103.09	0.2845	0.0636	75.90	4.8252	52.5	3.33745	48.3688
31	130.91	0.254	0.0507	95.23	4.8252	52.5	2.66022	38.5539
32	162	0.2311	0.0419	115.03	4.8252	75	3.14595	45.5935

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1.291 1.31

test motor chart , in Sheet1

Instructions Sheet1

yellow box.

Block 3

resist wire milliohm	Rm resistense Delta	Rm resist Star
3.18	2.12	6.37
3.84	2.56	7.67
4.49	2.99	8.98
5.14	3.43	10.28
5.79	3.86	11.58
6.44	4.30	12.89
7.10	4.73	14.19
7.75	5.17	15.50
8.40	5.60	16.80
9.05	6.03	18.10
9.70	6.47	19.41
10.36	6.90	20.71
11.01	7.34	22.02
11.66	7.77	23.32
12.31	8.21	24.62
12.96	8.64	25.93
13.62	9.08	27.23
14.27	9.51	28.54
14.92	9.95	29.84
15.57	10.38	31.15
16.22	10.82	32.45
16.88	11.25	33.75
17.53	11.69	35.06
18.18	12.12	36.36
18.83	12.56	37.67
19.48	12.99	38.97
20.14	13.42	40.27
20.79	13.86	41.58
21.44	14.29	42.88
22.09	14.73	44.19
22.75	15.16	45.49
23.40	15.60	46.79
24.05	16.03	48.10
24.70	16.47	49.40
25.35	16.90	50.71
26.01	17.34	52.01
26.66	17.77	53.31
27.31	18.21	54.62
27.96	18.64	55.92
28.61	19.08	57.23
29.27	19.51	58.53

LRK DELTA	LRK STAR	Wire Length (in)	resist wire milliohm	Rm resistense Delta	Rm resistense Star
18732	10815	7.5	6.37	4.24	12.73
9366	5407	9.1	7.67	5.11	15.34
6244	3605	10.6	8.98	5.98	17.95
4683	2704	12.2	10.28	6.85	20.56
3746	2163	13.7	11.58	7.72	23.17
3122	1802	15.3	12.89	8.59	25.78
2676	1545	16.8	14.19	9.46	28.38
2341	1352	18.4	15.50	10.33	30.99
2081	1202	19.9	16.80	11.20	33.60
1873	1081	21.5	18.10	12.07	36.21
1703	983	23.0	19.41	12.94	38.82
1561	901	24.5	20.71	13.81	41.42
1441	832	26.1	22.02	14.68	44.03
1338	772	27.6	23.32	15.55	46.64
1249	721	29.2	24.62	16.42	49.25
1171	676	30.7	25.93	17.29	51.86
1102	636	32.3	27.23	18.16	54.47
1041	601	33.8	28.54	19.02	57.07
986	569	35.4	29.84	19.89	59.68
937	541	36.9	31.15	20.76	62.29
892	515	38.5	32.45	21.63	64.90
851	492	40.0	33.75	22.50	67.51
814	470	41.5	35.06	23.37	70.11
780	451	43.1	36.36	24.24	72.72
749	433	44.6	37.67	25.11	75.33
720	416	46.2	38.97	25.98	77.94
694	401	47.7	40.27	26.85	80.55
669	386	49.3	41.58	27.72	83.16
646	373	50.8	42.88	28.59	85.76
624	360	52.4	44.19	29.46	88.37
604	349	53.9	45.49	30.33	90.98
585	338	55.5	46.79	31.20	93.59
568	328	57.0	48.10	32.07	96.20
551	318	58.5	49.40	32.94	98.81
535	309	60.1	50.71	33.80	101.41
520	300	61.6	52.01	34.67	104.02
506	292	63.2	53.31	35.54	106.63
493	285	64.7	54.62	36.41	109.24
480	277	66.3	55.92	37.28	111.85
468	270	67.8	57.23	38.15	114.45
457	264	69.4	58.53	39.02	117.06

Max turns

Block 5

mm	Ft Milliohms	D mm	Surface MM2	Turns max	mm2	Real Turns	Turns mm2	two wire R.
2	1.672	2.06	3.33292	1.2	4.05882	2	6.66585	0.836
1.9	1.853	1.96	3.01719	1.3	4.05882	3	9.05158	0.9265
1.8	2.065	1.86	2.71717	1.5	4.05882	6	16.303	1.0325
1.7	2.315	1.76	2.43286	1.7	4.05882	5.5	13.3807	1.1575
1.6	2.613	1.66	2.16425	1.9	4.05882	7	15.1497	1.3065
1.5	2.973	1.56	1.91135	2.1	4.05882	8	15.2908	1.4865
1.4	3.413	1.46	1.67416	2.4	4.05882	8	13.3933	1.7065
1.3	3.956	1.36	1.45268	2.8	4.05882	11	15.9794	1.978
1.2	4.65	1.257	1.24097	3.3	4.05882	10	12.4097	2.325
1.1	5.53	1.157	1.05137	3.9	4.05882	15	15.7706	2.765
1	6.69	1.057	0.87749	4.6	4.05882	18	15.7948	3.345
0.95	7.413	1.07	0.8992	4.5	4.05882	18	16.1857	3.7065
0.9	8.259	0.957	0.71931	5.6	4.05882	18	12.9475	4.1295

D factor 9834 Kv/Turn
Star/Y fact 5678 Kv/Turn
Term. Len **6** inch
Length per 3.09090909 inch

Notes:

Test Motor data

Nombre del Motor.						
Motor wind	Motor Kv.	No Load Current	Motor Resistance	I Max	P Max.	
	750					
Outside Diameter	Body Length	Total Shaft length	Shaft diameter	Motor weight		
				55		
Prpeller Manufacturer	Propeller Diameter	Propeller Pitch	Imput Voltage	Motor Amper.	Watts Imput.	Prop. RPM.
APC SF	10	4.7	11	10	110	6500
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0

Temperature test.

Propeller Secons		Propeller					
		Temp		Temp		Temp	
		Watts	Temp. F	Watts	Temp. F	Watts	Temp. F
	30						
	60						
	90						
	120						
	150						
	180						
	210						
	240						
	270						
	300						
	330						
	360						
	390						
	420						
	450						

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